

Cancer Incidence Profile in Sub-Saharan African-Born Blacks in the United States: Similarities and Differences With US-Born Non-Hispanic Blacks

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BACKGROUND: Sub-Saharan African-born blacks (ABs) are one of the fastest-growing populations in the United States. However, to the authors' knowledge, data regarding the cancer burden in this group are lacking, which would inform targeted cancer prevention and control. **METHODS:** The authors calculated age-standardized proportional incidence ratios (PIRs) comparing the frequency of the top 15 cancers in ABs with that of US-born non-Hispanic blacks (USBs) by sex and region of birth using incidence data for 2000 through 2012 from the Surveillance, Epidemiology, and End Results (SEER 17) program. **RESULTS:** Compared with USBs, ABs had significantly higher PIRs of infection-related cancers (liver, stomach, and Kaposi sarcoma), blood cancers (leukemia and non-Hodgkin lymphoma), prostate cancer, and thyroid cancers (females only). For example, the PIR for Kaposi sarcoma in AB versus USB women was 12.06 (95% confidence interval [95% CI], 5.23-18.90). In contrast, ABs had lower PIRs for smoking-related and colorectal cancers (eg, for lung cancer among men, the PIR was 0.30 [95% CI, 0.27-0.34]). Furthermore, cancer occurrence in ABs versus USBs varied by region of birth. For example, the higher PIRs for liver cancer noted among male ABs (PIR, 3.57; 95% CI, 1.79-5.35) and for thyroid cancer in female ABs (PIR, 3.03; 95% CI, 2.03-4.02) were confined to Eastern African-born blacks, whereas the higher PIR for prostate cancer (PIR, 1.90; 95% CI, 1.78, 2.02) was confined to Western African-born blacks. **CONCLUSIONS:** The cancer incidence profile of ABs is different from that of USBs and varies by region of birth, suggesting differences in environmental, cultural, social, and genetic factors. The findings of the current study could stimulate etiologic research and help to inform targeted interventions. *Cancer* 2017;123:3116-24. © 2017 American Cancer Society.

KEYWORDS: Africa, blacks, non-Hispanic, proportional incidence ratio, sub-Saharan.

INTRODUCTION

According to data from the 2015 American Community Survey, the number of black African immigrants (a majority of them from sub-Saharan Africa) in the United States increased from 1.4 million in 2006 to 2.1 million in 2015.¹ This number is expected to grow because of the Immigration Act of 1990, an act that was passed² to encourage immigration from underrepresented nations, as well as economic migration and refugee resettlements resulting from civil unrest (eg, Southern Sudan).

There are differences between sub-Saharan African-born blacks and those born in the United States with respect to culture, beliefs, socioeconomic status, dietary patterns, and genetics that could affect the occurrence of cancer and other diseases.³⁻⁷ Despite these differences, cancer occurrence for blacks in the United States is typically presented in aggregate without regard to country or region of birth, thereby masking important potential differences that can inform cancer prevention and control policies and programs for African-born black immigrants.⁸ In this article, we present what to our knowledge is for the first time the distribution of cancer incidence in sub-Saharan African-born blacks residing in the United States using data from the Surveillance, Epidemiology, and End Results (SEER) program.

MATERIALS AND METHODS

Data Description

We used the SEER program registries, representing approximately 28% of the US population,⁹ to identify cancer cases diagnosed from 2000 through 2012. We limited our analysis to cases with black race and non-Hispanic ethnicity with a known place of birth, excluding approximately 38% of the total number of black cases (207,498 cases). Data from 17 of

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the 18 SEER registries were used because the Alaska Native registry did not contain any black cases. Hispanic blacks were excluded because they represent a small percentage of blacks in the United States and may have different cultural practices. We further limited our analysis to black patients diagnosed with 1 of the 15 most common cancers in males and females in Africa according to GLOBOCAN 2012.¹⁰ These included cancers of the lip and oral cavity, esophagus, stomach, liver, colorectum, lung and bronchus, and urinary bladder as well as Kaposi sarcoma, non-Hodgkin lymphoma (NHL), and leukemia in both men and women; cancers of prostate, larynx, nasopharynx, other pharynx, and pancreas in men; and cancers of the cervix uteri, breast, corpus uteri, ovary, and thyroid in females. We chose to use the top 15 cancers in Africa instead of the top 15 cancers in US-born blacks because in general cancer patterns in first-generation immigrants are known to be similar to those of their countries of origin rather than to their host countries.^{11,12}

We obtained place of birth in the SEER database through a custom data request because it is suppressed in the publicly available database (Mandi Yu, personal communication, October 2015). We categorized birthplace into sub-Saharan African-born blacks (ABs) and US-born non-Hispanic blacks (USBs). Place of birth for ABs was categorized further according to the United Nations region classification into Eastern (Burundi, Djibouti, East Africa-not otherwise specified [NOS], Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mauritius, Mozambique, Rwanda, Somalia, United Republic of Tanzania, Uganda, Zambia, and Zimbabwe), Western (Gambia, Ghana, Liberia, Niger, Nigeria, Senegal, Sierra Leone, Togo, and West Africa-NOS), Southern (Southern Africa-NOS, South Africa, and Swaziland), and Middle African (Angola, Cameroon, and Democratic Republic of the Congo) regions.¹³ We identified 4465 newly diagnosed cases from 1 of the top 15 cancer types among ABs in the SEER database during the study period. Of these cases, 1682 were from Eastern Africa, 2064 were from Western Africa, 104 were from Southern Africa, 43 were from Middle Africa, and 572 were from other African countries NOS. During the same time period, we identified 327,773 USBs.

We considered several covariates in the analysis. Age at the time of diagnosis was dichotomized into <20 years, 20 to 39 years, 40 to 59 years, 60 to 69 years, or ≥70 years; the median county-level household income was categorized according to the 2000 US Census national median family income into <\$45,200 and ≥\$45,200; and marital status was categorized into married (which included common-law partner) and unmarried (which included

never married, separated, divorced, and widowed). Health insurance, which was available for cases included for 2007 through 2012 only, was categorized into insured or uninsured for the age groups of <65 years and ≥65 years separately because of substantial differences in the percentage of uninsured individuals between the 2 age categories. For cancers with early detection methods/tests (female breast, prostate, cervix, and colorectum), stage of disease at the time of diagnosis was categorized into localized, regional, distant, or unknown/missing according to SEER summary stage 2000.⁹

Statistical Analyses

We compared age, sex, and socioeconomic status (marital status, county-level median income, and health insurance status) between ABs and USBs using the Pearson chi-square test. To determine whether the distributions of cancer types varied between ABs and USBs, age-adjusted proportional incidence ratios (PIRs) for each of the 15 cancer types were estimated with corresponding 95% confidence intervals (95% CIs) using a method previously described.¹⁴ Briefly, the PIR was calculated as the observed-to-expected ratio. The expected number of cases among ABs for each cancer type was calculated by multiplying the total number of observed cancers in an age group (aged birth to 4 years, 5-9 years, . . . , aged ≥85 years) by the corresponding percentage of cases in USBs, which was considered the standard. These observed and expected numbers then were summed over age groups to obtain the overall age-adjusted observed and expected numbers, respectively. For select cancers, we also calculated PIRs for Eastern African-born blacks and Western African-born blacks versus USBs. The PIRs for the other African regions were not calculated because of sparse numbers. Comparisons based on <10 cases in ABs or USBs were excluded from the PIR analysis because estimates were deemed unstable. Because detailed population data by age and sex for black immigrants from sub-Saharan Africa were not available from the US Census Bureau, we could not calculate incidence rates, and instead used PIRs to describe and compare the burden of cancer between ABs and USBs.

Using multinomial logistic regression models adjusting for age, county-level household income, and sex (colorectal cancer only), we compared the odds of being diagnosed at a distant or regional stage of disease compared with a localized stage (base category) between ABs and USBs for cancers with early detection methods. An interaction between sex and place of birth also was assessed to determine whether the association between

stage at diagnosis and place of birth varied between males and females. Cases with unknown/missing stage at diagnosis (among ABs: 20 cases for female breast, 32 cases for prostate, 7 cases for cervix, and 14 cases for colorectal and among USBs: 1448 cases for female breast, 2745 cases for prostate, 259 cases for cervix, and 2738 cases for colorectal) were excluded from analyses.

To assess the influence of missing place of birth on the study findings, we compared sociodemographic characteristics and the distribution of cancer among black patients with an unknown and known place of birth. We also conducted models examining stage of disease at the time of diagnosis including missing place of birth as a covariate. Statistical analyses were performed using Stata statistical software (14.0.1 MP; StataCorp LP, College Station, Tex) using 2-tailed tests with a significance level set at an $\alpha < .05$.

RESULTS

Demographic and area-based socioeconomic status varied significantly between newly diagnosed AB and USB patients with cancer (Table 1). Compared with USBs, ABs were younger (median age at the time of diagnosis: 55 years vs 64 years) and were more likely to be married (56% vs 36%) and uninsured, both among those aged <65 years (15% vs 9%) and those aged ≥ 65 years (18% vs 1%). ABs were more likely to reside in more affluent counties compared with USBs (60% vs 33%). There were no significant differences noted with regard to stage of disease at the time of diagnosis for cervical, breast, and colorectal cancers between ABs and USBs (see Supporting Information Table 1).

Table 2 shows the relative frequency of the top 15 cancers in ABs compared with USBs by sex. Among AB females, breast cancer was the most commonly diagnosed cancer, accounting for 35.0% of the total cases (741 of 2119 cases), followed by colorectal (6.0%) and thyroid (4.9%) cancer. The pattern was similar for USB females except that lung cancer replaced thyroid cancer as the third most commonly diagnosed cancer. Among AB males, prostate cancer was the most commonly diagnosed cancer, representing approximately 37.9% of cases (888 of 2346 cases), followed by colorectal (8.3%) and liver (7.2%) cancer. Prostate cancer also was the most frequently diagnosed cancer in USB males (28.6% of the total cases), but was followed by lung (19.0%) and colorectal (10.7%) cancers. It is interesting to note that lung cancer accounted for 19% of all cancer diagnoses among USB males, but only 5% in AB males.

Figure 1 shows PIRs for the 15 top cancers in ABs compared with USBs by sex. The PIR was significantly higher in ABs for 4 of the 15 top cancers (liver, stomach, Kaposi sarcoma, and leukemia) in both females and males, and for an additional 2 sites (prostate cancer and NHL) in males. In contrast, the PIR was significantly lower in ABs for colorectal and lung cancer in both males and females and for cancers of urinary bladder, larynx, and lip and oral cavity in males (Table 2) (Figs. 1A and 1B).

Table 3 depicts the PIRs comparing ABs versus USBs by region (Eastern African-born and Western African-born blacks) for 5 select cancers by sex. The significantly higher PIRs noted for thyroid cancer in AB females and for liver cancer and NHL in AB males were observed only in Eastern African-born blacks, whereas a higher PIR for prostate cancer was confined to Western African-born blacks.

The percentage of blacks for whom data regarding place of birth were missing was greater among younger patients and those diagnosed in more recent years and varied across the registry (see Supporting Information Table 2). In addition, there were disproportionate numbers of breast, thyroid (in females), and prostate cancer cases with a missing place of birth (see Supporting Information Table 3). The inclusion of missing place of birth in models predicting stage of disease did not alter the point estimate for ABs, and those with an unknown place of birth had lower odds of distant stage disease (see Supporting Information Table 4).

DISCUSSION

In the current study, to the best of our knowledge, we provide the first comprehensive description of the cancer incidence profile of ABs in the United States. Compared with USBs, ABs had significantly higher PIRs for infection-related cancers (liver, stomach, and Kaposi sarcoma) and blood cancers (leukemia and NHL), but lower PIRs for smoking-related cancers (lung, lip and oral cavity, larynx, and urinary bladder). ABs also had significantly higher PIRs for thyroid cancer in females and for prostate cancer. It is interesting to note that the significantly higher PIRs for thyroid cancer in females and for liver cancer and NHL in males were confined to Eastern African-born blacks, whereas the higher PIR for prostate cancer was observed only in Western African-born blacks.

Infectious agents are the causes of some of the most commonly diagnosed cancers in Africa.¹⁵ The higher PIRs of infection-associated cancers in ABs were not unexpected considering the high prevalence of infectious agents associated with cancer such as herpesvirus 8 (Kaposi

TABLE 1. Case Characteristics Among Sub-Saharan African-Born and US-Born Blacks: SEER 17, 2000 Through 2012

Categories	Sub-Saharan Born		P ^a	African Regions Among Sub-Saharan Blacks				
	N = 4465	US Born N = 327,773		Eastern N = 1682	Western N = 2064	Southern N = 104	Middle N = 43	Other ^b N = 572
	% (No.)	% (No.)		% (No.)	% (No.)	% (No.)	% (No.)	% (No.)
Age at diagnosis, y								
<20	1.4 (61)	1.0 (3324)	<.001	1.6 (27)	1.1 (23)	1.0 (1)	2.3 (1)	1.6 (9)
20-39	13.4 (599)	4.1 (13,576)		16.9 (284)	10.6 (218)	7.7 (8)	23.3 (10)	13.8 (79)
40-59	47.3 (2113)	32.4 (106,345)		44.6 (750)	49.2 (1016)	48.1 (50)	46.5 (20)	48.4 (277)
60-69	23.1 (1031)	26.9 (88,231)		20.9 (351)	24.8 (512)	27.9 (29)	16.3 (7)	23.1 (132)
≥70	14.8 (661)	35.5 (116,297)		16.1 (270)	14.3 (295)	15.4 (16)	11.6 (5)	13.1 (75)
Sex								
Male	52.5 (2346)	52.2 (171,131)	.659	41.8 (703)	59.8 (1235)	50.0 (52)	65.1 (28)	57.3 (328)
Female	47.5 (2119)	47.8 (156,642)		58.2 (979)	40.2 (829)	50.0 (52)	34.9 (15)	42.7 (244)
Marital status								
Married	56.1 (2504)	36.4 (119,469)	<.001	52.1 (877)	60.2 (1241)	53.9 (56)	41.9 (18)	54.5 (312)
Not married	37.7 (1682)	58.3 (190,963)		40.2 (677)	34.5 (713)	39.4 (41)	53.4 (23)	39.9 (228)
Unknown	6.2 (279)	5.3 (17,341)		7.7 (128)	5.3 (110)	6.7 (7)	4.7 (2)	5.6 (32)
Health insurance								
at age <65 y ^c								
Insured	81.6 (1191)	86.9 (68,685)	<.001	86.5 (477)	78.4 (541)	90.2 (37)	88.9 (8)	75.7 (128)
Uninsured	15.1 (221)	8.9 (7064)		11.3 (62)	18.1 (125)	7.3 (3)	11.1 (1)	17.8 (30)
Unknown	3.3 (48)	4.2 (3351)		2.2 (12)	3.5 (24)	2.5 (1)	0.0 (0)	6.5 (11)
Health insurance								
at age ≥65 y ^c								
Insured	77.3 (438)	92.1 (64,976)	<.001	82.8 (183)	71.4 (185)	90.2 (11)	50.0 (1)	79.4 (58)
Uninsured	17.5 (99)	0.7 (522)		12.2 (27)	23.9 (62)	8.3 (1)	50.0 (1)	11.0 (8)
Unknown	5.3 (30)	7.2 (5087)		5.0 (11)	4.7 (12)	0.0 (0)	0.0 (0)	9.6 (7)
County-level								
median								
household								
income								
<\$45,200	40.4 (1803)	66.7 (218,522)		30.1 (507)	47.2 (975)	53.8 (56)	55.8 (43)	42.1 (241)
≥\$45,200	59.6 (2662)	33.3 (109,237)	<.001	69.9 (1175)	52.8 (1089)	46.2 (48)	44.2 (19)	57.9 (331)
Median age at								
diagnosis, y								
	54.5 (45.0-64.5)	63.5 (53.0-73.5)		54.0 (43.0-64.0)	55.5 (47.0-64.0)	56.5 (48.0-66.0)	51.5 (40.0-60.5)	53.5 (44.0-64.0)

Abbreviations: IQR, interquartile range; SEER, Surveillance, Epidemiology, and End Results.

^aComparison between Sub-Saharan-born and US-born non-Hispanic black individuals.^bAfrican countries not otherwise specified.^c2007 through 2012.

TABLE 2. Relative Frequency and PIR of the Top 15 Cancers in Sub-Saharan African-Born Blacks Compared With US-Born Blacks: SEER 17, 2000 Through 2012

	Sub-Saharan African-Born Blacks % (No.)	US-Born Non-Hispanic Blacks % (No.)	PIR (95% CI)
Females			
Breast	35.0 (741)	29.3 (45,832)	1.01 (0.93-1.09)
Colorectum	6.0 (128)	11.4 (17,893)	0.66 (0.52-0.80) ^a
Thyroid	4.9 (103)	1.5 (2318)	2.05 (1.47-2.64) ^a
Lung and bronchus	4.4 (94)	14.3 (22,331)	0.38 (0.29-0.48) ^a
Cervix uteri	3.9 (83)	2.3 (3631)	1.09 (0.39-1.81)
Corpus uteri	3.8 (80)	4.4 (6862)	0.92 (0.61-1.25)
NHL	3.4 (72)	2.5 (3865)	1.19 (0.93-1.46)
Leukemia	3.0 (64)	1.9 (3039)	1.62 (1.21-2.03) ^a
Stomach	2.6 (54)	2.1 (3339)	1.54 (1.05-2.04) ^a
Liver	2.6 (54)	1.0 (1600)	2.76 (1.20-4.33) ^a
Ovary	2.5 (53)	2.5 (3968)	0.96 (0.65-1.29)
Lip and oral cavity	0.9 (19)	0.9 (1399)	0.90 (0.43-1.37)
Urinary bladder	0.8 (17)	1.6 (2500)	0.84 (0.45-1.23)
Esophagus	0.7 (14)	0.9 (1365)	0.90 (0.39-1.40)
Kaposi sarcoma	0.7 (14)	0.0 (49)	12.06 (5.23-18.90) ^a
All other cancers	25.0 (529)	23.4 (36,646)	
Males			
Prostate	37.9 (888)	28.6 (48,858)	1.53 (1.44-1.61) ^a
Colorectum	8.3 (195)	10.7 (18,252)	0.81 (0.65-0.97) ^a
Liver	7.2 (169)	3.0 (5099)	2.15 (1.67-2.64) ^a
NHL	5.9 (139)	3.0 (5083)	1.34 (1.20-1.48) ^a
Lung and bronchus	5.0 (121)	19.0 (32,480)	0.30 (0.27-0.34) ^a
Leukemia	3.0 (70)	2.2 (3679)	1.40 (1.21-1.60) ^a
Stomach	2.3 (55)	2.1 (3675)	1.14 (1.07-1.22) ^a
Pancreas	2.3 (53)	3.2 (5550)	0.73 (0.42-1.05)
Urinary bladder	1.9 (44)	2.8 (4770)	0.76 (0.67-0.86) ^a
Kaposi sarcoma	1.6 (37)	0.5 (795)	1.56 (1.07-2.04) ^a
Larynx	0.9 (22)	1.9 (3273)	0.49 (0.32-0.66) ^a
Lip and oral cavity	0.8 (18)	1.6 (2790)	0.39 (0.12-0.66) ^a
Nasopharynx	0.6 (13)	0.2 (371)	1.64 (0.60-2.68)
Esophagus	0.4 (9)	1.9 (3325)	
Other pharynx	0.2 (4)	0.9 (1448)	
All other cancers	22.4 (527)	18.5 (31,683)	

Abbreviations: 95% CI, 95% confidence interval; NHL, non-Hodgkin lymphoma; PIR, proportional incidence ratio; SEER, Surveillance, Epidemiology, and End Results.

A PIR >1.00 indicates a higher proportional incidence compared with US-born blacks whereas a PIR <1.00 indicates a lower proportional incidence compared with US-born blacks.

^aValue is statistically significant.

sarcoma), secondary to human immunodeficiency virus (HIV)/acquired immune deficiency syndrome (AIDS); hepatitis B and C viruses (hepatocellular carcinoma); and *Helicobacter pylori* (gastric cancer) in both African indigenous populations and immigrants in the United States.¹⁶ For example, the prevalence of chronic hepatitis B virus infection was >8% in blacks born in sub-Saharan Africa compared with <2% in US-born blacks.^{17,18} It also is possible that a substantial percentage of ABs may have been exposed to aflatoxins, a known risk factor for liver cancer, before migrating to the United States through the consumption of contaminated food.¹⁹ However, the higher PIR for liver cancer observed in ABs was confined to Eastern African-born blacks, which in part reflects differences in the prevalence of hepatitis virus infection by

region of birth among ABs.^{20,21} The higher PIR noted for Kaposi sarcoma in ABs was likely secondary to HIV/AIDS, because it was estimated that HIV diagnosis rates among African-born US residents were 6 times higher than those of the US population.^{22,23} Furthermore, the substantially greater PIR for Kaposi sarcoma observed among women compared with men reflects the very low percentage of the disease in USB women.

The lower PIR of smoking-related cancers in ABs compared with USBs reflects the historical low prevalence of smoking in African-born blacks. Between 1990 and 1994, the prevalence of smoking in foreign-born blacks (14%) was significantly lower than that in the USBs (30%).²⁴ Africa has by far the lowest smoking-related cancer mortality worldwide. According to the World Health

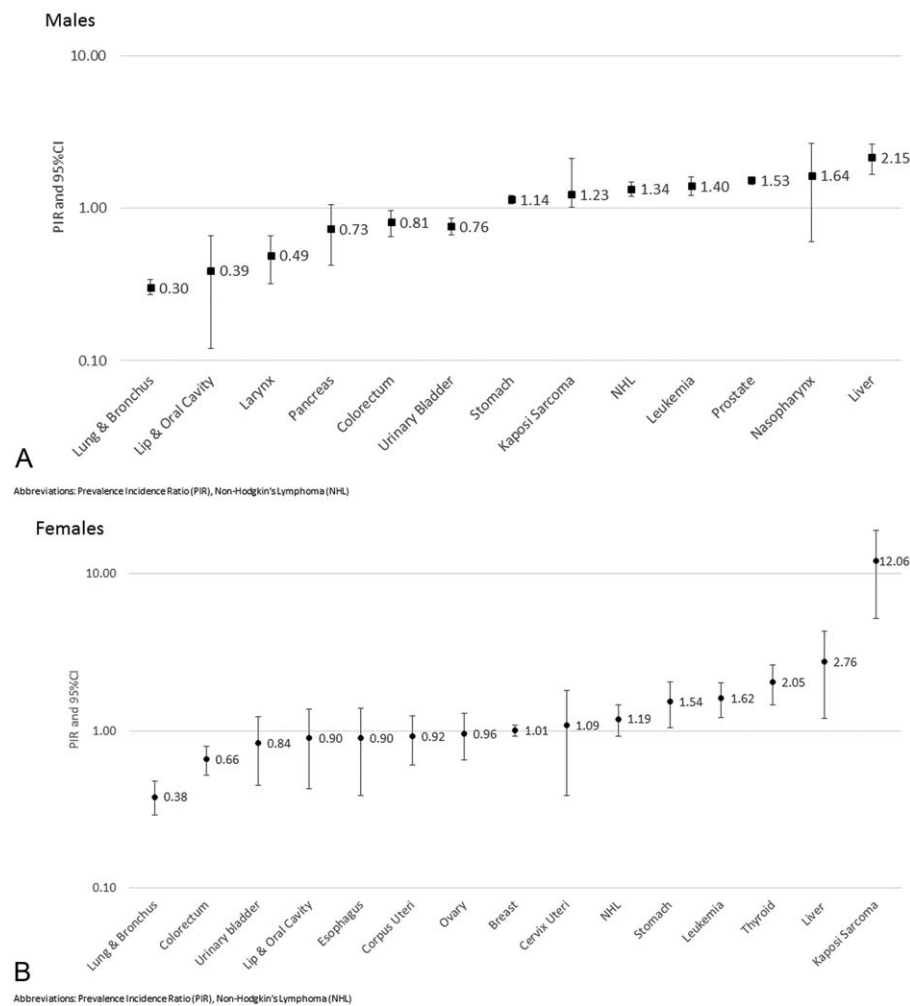


Figure 1. (A) Proportional incidence ratios (PIRs) of the top 15 cancers in blacks born in sub-Saharan Africa compared with US-born blacks for males: Surveillance, Epidemiology, and End Results (SEER) 17 data, 2000 through 2012. (B) PIRs of the top 15 cancers in blacks born in sub-Saharan Africa compared with US-born blacks for females: SEER 17 data, 2000 through 2012. 95% CI indicates 95% confidence interval; NHL, non-Hodgkin lymphoma.

Organization, approximately 13% of cancer deaths in men and 1% of cancer deaths in women in Africa in 2004 were attributed to tobacco use, compared with 28% in men and 4% in women in the Mediterranean, which has the second lowest smoking attributable fractions, and 33% in men and 18% in women in the Americas.²⁵

Similar to USB women, breast cancer is the most frequently diagnosed cancer in AB women, which is not surprising because it also is the most commonly diagnosed cancer in most parts of sub-Saharan Africa.^{26,27} Breast cancer incidence rates have increased rapidly in sub-Saharan Africa over the last several decades, which may reflect changes in risk factors associated with economic development, including delayed child bearing, decreased

parity, shortened duration of breastfeeding, and increased prevalence of obesity, as well as an increased awareness of breast cancer.²⁸⁻³¹

Prostate cancer was the most frequently diagnosed cancer in AB men, which is consistent with data among black men born in sub-Saharan Africa and USB men.^{26,27} Furthermore, men of African descent originating from Western Africa are known to have the highest rates of prostate cancer worldwide.³² The higher PIR for prostate cancer in Western African-born blacks (65% of whom were identified as Nigerian-born individuals) compared with USBs and in USBs than Eastern African-born blacks (60% of whom were identified as Ethiopian-born or Eritrean-born individuals) suggest shared genetic risk

TABLE 3. PIR of the 5 Most Frequent Cancers in Sub-Saharan African-Born Black Migrants Compared With US-Born Blacks By Region of Origin: SEER 17, 2000 Through 2012

	Western African-Born Blacks		Eastern African-Born Blacks			
	Percent	Cases	PIR (95% CI)	Percent	Cases	PIR (95% CI)
Female ^a						
Breast	35.5	294	1.04 (0.93-1.15)	34.0	333	0.98 (0.85-1.12)
Colorectum	7.0	58	0.74 (0.51-0.97) ^b	5.1	50	0.57 (0.41-0.72) ^b
Thyroid	2.7	22	1.19 (0.65-1.74)	7.5	73	3.03 (2.03-4.02) ^b
Lung and bronchus	4.0	33	0.33 (0.16-0.50) ^b	4.6	45	0.41 (0.27-0.55) ^b
Corpus uteri	4.0	33	0.96 (0.49-1.43)	2.9	28	0.71 (0.41-1.01)
Male ^c						
Prostate	48.5	599	1.90 (1.78-2.02) ^b	20.2	142	0.84 (0.76-0.92) ^b
Colorectum	8.2	101	0.80 (0.62-0.98) ^b	9.2	65	0.90 (0.65-1.15)
Liver	4.9	61	1.40 (0.11-2.69)	11.0	77	3.57 (1.79-5.35) ^b
NHL	4.8	59	1.14 (0.89-1.40)	7.5	53	1.59 (1.36-1.82) ^b
Lung and bronchus	4.0	49	0.23 (0.18-0.27) ^b	7.0	49	0.42 (0.33-0.51) ^b
Both sexes ^d						
Colorectum	7.7	159	0.78 (0.65-0.91) ^b	6.8	115	0.71 (0.58-0.84) ^b
Liver	3.3	68	1.48 (0.27-2.69)	7.1	119	3.52 (1.69-5.36) ^b
Lung and bronchus	4.0	82	0.27 (0.21-0.33) ^b	5.6	94	0.40 (0.32-0.49) ^b
NHL	4.0	83	1.17 (0.96-1.40)	5.5	93	1.44 (1.22-1.66) ^b
Thyroid	1.7	35	1.25 (0.69-1.80)	5.1	86	3.25 (2.02-4.47) ^b

Abbreviations: 95% CI, 95% confidence interval; NHL, non-Hodgkin lymphoma; PIR, proportional incidence ratio; SEER, Surveillance, Epidemiology, and End Results.

A PIR >1.00 indicates a higher proportional incidence compared with US-born blacks, whereas a PIR <1.00 indicates a lower proportional incidence compared with US-born blacks.

^a Percentages are based on 829 Western African-born and 979 Eastern African-born female blacks, respectively.

^b Value is statistically significant.

^c Percentages are based on 1235 Western African-born and 703 Eastern African-born male blacks, respectively.

^d Percentages are based on 2064 Western African-born and 1682 Eastern African-born male and female blacks, respectively.

factors^{33,34} because the majority of USBs trace their roots to Western Africa because of the involuntary migration to North America as early as the 16th century.³⁴

The incidence of colorectal cancer globally is highest in economically developed countries in part because of the high prevalence of obesity and the consumption of an unhealthy diet in these countries.^{35,36} The lower colorectal cancer PIR in ABs compared with USBs is consistent with the global patterns.^{35,36} The lower risk of colorectal cancer noted among native Africans has been attributed to a lower prevalence of red meat and animal fat consumption and differences in the colonic microbiome, including lower levels of hydrogen-producing and secondary bile salt-producing bacteria.³⁷ O'Keefe et al demonstrated reciprocal changes in markers of inflammation and proliferation in the colonic mucosa within 2 weeks of a closely supervised food exchange between rural Africans and African Americans.³⁸

Similar to liver cancer, the higher PIRs noted among ABs compared with USBs for thyroid cancer and NHL were confined to Eastern African-born blacks. Reasons for these patterns are unknown, but may reflect regional differences in putative risk factors for these cancers. For

example, the highest incidence and mortality due to Epstein-Barr virus-associated NHL in Africa was reported in Eastern parts of the continent.³⁹ Similarly, the high incidence of thyroid cancer in parts of Eastern Africa has been attributed to a high prevalence of iodine deficiency-associated goiter in that region.⁴⁰⁻⁴²

There are several limitations to the current study. First, we did not have information regarding the length of stay in the United States, which could affect health behaviors and access to care.⁸ Specifically, although acculturation appears to have benefits in terms of a better understanding of the importance of cancer screening and an awareness of health services, it also could contribute to increased exposure to cancer risk factors such as smoking, physical inactivity, and unhealthy dietary habits. In addition, we could not calculate cancer incidence rates because of a lack of reliable population estimates for ABs, thereby underscoring the need for a concerted effort by the US Census Bureau to collect detailed population estimates for black immigrants from Africa and other parts of the world. Such data are important to assess the effect of the increasing size of the migrant population on the national cancer rates and trends among black individuals as first-

generation immigrants tend to have a lower overall cancer incidence.^{43,44}

Moreover, the findings of the current study should be interpreted with caution because approximately 38% of non-Hispanic blacks were excluded from our main analyses due to having an unknown birthplace in the SEER database. As with previous studies using the SEER database for the analysis of cancer patterns by birthplace,⁴⁵⁻⁴⁸ completeness of birthplace was not random and varied substantially by age, median family income, year of diagnosis, registries, and cancer types (see Supporting Information Tables 2 and 3). In addition, previous studies reported that birthplace information in the SEER database was more likely to be recorded for foreign-born, less acculturated, and dead patients.⁴⁵⁻⁴⁸ However, despite these limitations, it is interesting to note that when birthplace information is recorded in the registry database it is highly accurate compared with self-reported data,^{47,48} and imputation data reasonably predict unknown birthplace according to US birth/foreign birth for some ethnicities (eg, Hispanics), although not according to region or country of birth.⁴⁹ Furthermore, the inclusion of an unknown place of birth into models predicting stage of disease at the time of diagnosis did not appear to alter the main findings of the current study.

The cancer incidence profile of ABs is different from that of USBs and varies by countries (region) of origin, suggesting regional differences in environmental, cultural, social, and genetic factors. The current study findings could stimulate etiologic research and help to inform targeted cancer control interventions.

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CONFLICT OF INTEREST DISCLOSURES

The authors made no disclosures.

AUTHOR CONTRIBUTIONS

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